



Day 4: OSI MODEL (Open Systems Interconnection Model)

The OSI Model is a conceptual framework used to understand and standardize how different networking protocols and technologies communicate with each other. It was developed by the International Organization for Standardization (ISO) in 1984.

The model divides network communication into 7 layers. Each layer serves a specific function and only communicates with the layers directly above and below it.

THE 7 LAYERS OF THE OSI MODEL (Top to Bottom)

LAYER 7: APPLICATION LAYER

Definition: The topmost layer that provides network services directly to the end-user's applications. It is the interface between the software you use and the network.

Function: Handles resource sharing, file transfers, email services, and remote access. It identifies communication partners and determines resource availability.

Protocols/Examples: HTTP/HTTPS (Web browsing), FTP (File transfers), SMTP/POP3/IMAP (Email), DNS (Domain Name System), Telnet/SSH (Remote access).

Real-World Analogy: The actual browser window (like Chrome or Firefox) that you type your search into.

LAYER 6: PRESENTATION LAYER

Definition: The "translator" layer that ensures data sent from one system's Application Layer can be read by another system's Application Layer.

Function: Handles data formatting, encryption/decryption, and compression/decompression so that the data is presented in a standard format (e.g., converting EBCDIC to ASCII).

Protocols/Examples: SSL/TLS (Encryption), JPEG/GIF/PNG (Image formatting), MPEG (Video compression), ASCII/EBCDIC (Character encoding).

Real-World Analogy: A translator who takes your English sentence, translates it into French, and encrypts it into a secret code before sending it.

LAYER 5: SESSION LAYER

Definition: Manages, establishes, maintains, and terminates communication sessions (connections) between two applications on different devices.

Function: Responsible for dialogue control (who speaks when - half-duplex or full-duplex) and synchronization (adding checkpoints to resume interrupted transfers).

Protocols/Examples: NetBIOS, RPC (Remote Procedure Call), PPTP (Point-to-Point Tunneling Protocol), SMB (Server Message Block).

Real-World Analogy: A telephone call moderator who dials the number, makes sure both people are connected, puts the call on hold if needed, and hangs up when done.

LAYER 4: TRANSPORT LAYER

Definition: Ensures complete, reliable end-to-end data transfer between the source and destination hosts. It segments data and reassembles it at the receiving end.

Function: Handles error detection, flow control, and data segmentation. It decides how much data to send, at what speed, and retransmits lost packets.

Protocols/Examples: TCP (Transmission Control Protocol - reliable, connection-oriented) and UDP (User Datagram Protocol - fast, connectionless).

Real-World Analogy: A shipping manager who tracks your package, splits heavy furniture into multiple boxes, labels each box with a tracking number, and reships any lost boxes.

LAYER 3: NETWORK LAYER

Definition: Handles routing and forwarding of data packets across different networks. It determines the best physical path for data to travel from source to destination.

Function: Logical addressing (IP addresses), packet forwarding, and routing (finding the shortest/best path using routers).

Protocols/Examples: IPv4, IPv6, ICMP (Ping), RIP, OSPF, BGP (Routing protocols), Routers (Hardware).

Real-World Analogy: A GPS navigation system or a postal sorting office that decides the best highway route for your package to get from New York to California.

LAYER 2: DATA LINK LAYER

Definition: Provides node-to-node data transfer between two directly connected devices on the same physical network segment.

Function: Packages raw bits from Layer 1 into frames. Handles error detection (CRC) and flow control on the local link. Uses MAC addresses (hardware addresses).

Sub-layers: LLC (Logical Link Control - identifies network protocols) and MAC (Media Access Control - controls hardware addressing and access to the physical medium).

Protocols/Examples: Ethernet, PPP (Point-to-Point Protocol), Wi-Fi (802.11), Switches (Hardware), Bridges.

Real-World Analogy: The local mailman who picks up all the letters from your neighborhood mailbox and delivers them to the main post office. He knows every house's physical address (MAC) on his street.

LAYER 1: PHYSICAL LAYER

Definition: The lowest layer that deals with the actual physical transmission of raw, unstructured data bits (0s and 1s) over a physical medium.

Function: Defines the electrical, mechanical, procedural, and functional specifications for the physical hardware (cables, voltages, pin layouts, signaling rates). It converts digital bits into electrical/radio/optical signals.

Protocols/Examples: Ethernet cables (RJ45, Cat5e/Cat6), Fiber optics, Coaxial cables, USB cables, Hubs, Repeaters, Bluetooth/Wi-Fi radio waves.

Real-World Analogy: The truck, airplane, or delivery van and the actual roads/runways used to physically carry the boxes from one location to another.

HOW DATA TRAVELS THROUGH THE OSI MODEL (Encapsulation)

When you send data (e.g., an email), it travels down the OSI model on your computer, and up the OSI model on the receiving computer. Each layer adds its own information (called a header) to the original data.

This process is called Encapsulation:

- ❖ Application (L7): User creates the data (e.g., "Hello").
- ❖ Presentation (L6): Data is encrypted/compressed.
- ❖ Session (L5): Checkpoints are added.

- ❖ Transport (L4): Data is chopped into Segments (TCP) or Datagrams (UDP). A port number is added (e.g., Port 25 for email).
- ❖ Network (L3): Segments are wrapped into Packets. A source and destination IP address is added (e.g., 192.168.1.1).
- ❖ Data Link (L2): Packets are wrapped into Frames. A source and destination MAC address is added (e.g., 00:1A:2B:3C:4D:5E).
- ❖ Physical (L1): Frames are converted into Bits (electrical pulses, light, or radio waves) and sent over the wire/air.

On the receiving side, the process is reversed (called De-encapsulation). Each layer strips off its corresponding header until the original data (Layer 7) is presented to the receiving application.